

Time-reversal inference with uncertain observation channels

Dev D. Goyal^{1,*}

¹*Independent researcher*

(Dated: 7 October 2026)

Time asymmetry in a recording can arise from the physical source or from unequal detector responses. We determine when these explanations can be distinguished for stationary Gaussian sources observed through linear channels. If source frequencies and channel delays are unrestricted, unequal delays can make a reversible source reproduce the joint probabilities of any finite stationary Gaussian record. This holds when the recorded fluctuations have no exact linear constraint. Source inference therefore requires information beyond the record. We derive two finite-record tests that use such information. A calibrated two-channel test bounds the asymmetry allowed by relative detector phase. A three-channel spectral product cancels detector phase under a sampled diagonal response model. Both tests control false rejection using a bound on dependence across all channels and recorded times. The spectral test also needs a bound on covariances beyond the retained time separations. In a simulated Brownian gyrator at equilibrium, unequal sensors produce raw bootstrap detections in 397 of 400 records of length $N = 524289$, while the calibrated source test rejects none. At the same length, the calibrated test detects all 200 records at a fourfold bath-temperature ratio but none at a twofold ratio. The tests thus separate detector-induced asymmetry from source evidence, with a substantial cost in recording length. A rotational example establishes the remaining physical limit: identical sampled laws can have unboundedly different continuous-time entropy-production rates.

I. INTRODUCTION

A particle at thermal equilibrium has no preferred direction of circulation. Yet two sensors that record its coordinates can produce a preferred time order if their responses differ. One sensor may respond immediately while the other retains a small contribution from an earlier position. The relation between the two recorded signals then depends on their time order, even though the particle remains at equilibrium (Fig. 1). More data make this recorded asymmetry easier to detect. They do not establish whether it belongs to the particle or to the measurement.

This distinction matters when fluctuations are used to identify physical activity. Detailed balance means that forward and reverse transitions balance in pairs. Violations have been detected in active biological systems [9]. Trajectory asymmetry has also been used to estimate entropy production in actomyosin, a network of interacting actin and myosin proteins [11]. Entropy production measures thermodynamic irreversibility. Current fluctuations and cycling frequencies provide related measures of nonequilibrium behavior [12–14]. These methods connect statistical time order to physical driving under specified dynamical and observation assumptions [1–4, 10]. Unequal observation channels introduce another possible source of that order. Fluorescence-filter switching delays measurements at different wavelengths [7]; photodiodes and electronics alter the temporal response of optical traps [8]. Cross spectra express how two signals vary together at each frequency. Their phase describes relative timing, which these detector effects can change [5, 6].

We ask when a finite recording rules out every reversible source allowed by the observation model. A reversible source has the same joint probabilities when time runs forward or backward. We study stationary Gaussian sources. Their statistical rules do not change with time, and their joint distributions are normal. Their means and covariances determine those distributions completely. This gives a precise setting in which to separate an ambiguity of the measurement from uncertainty caused by a finite sample. We allow stable linear responses that act separately on the channels, together with stationary Gaussian observation errors that are uncorrelated across channels and with the source.

The ambiguity is complete if source spectra and channel delays are unrestricted. We prove that any nonsingular finite stationary Gaussian law can arise from a reversible source observed through unequal delays. Thus no test with controlled false rejection over that class can distinguish an asymmetric alternative. The result does not depend on estimating the law poorly: it holds even when the finite-record probabilities are known exactly. Arbitrarily small suitable delays give the same obstruction when unrestricted continuous-time frequencies can alias under sampling.

Useful source inference must therefore restrict the possible measurement explanations. We develop two ways to do this. For two channels, calibration bounds the asymmetry that the relative response phase can create. With three channels, a product of pairwise cross spectra cancels detector phase under the sampled response model. Each operation constrains a quantity defined by the underlying probability law, called its population value. An estimate from one finite record can differ from that value. We then convert that constraint into a finite-record test by controlling dependent sampling errors. The required

* dvgy@outlook.com; ORCID: 0009-0009-9132-744X

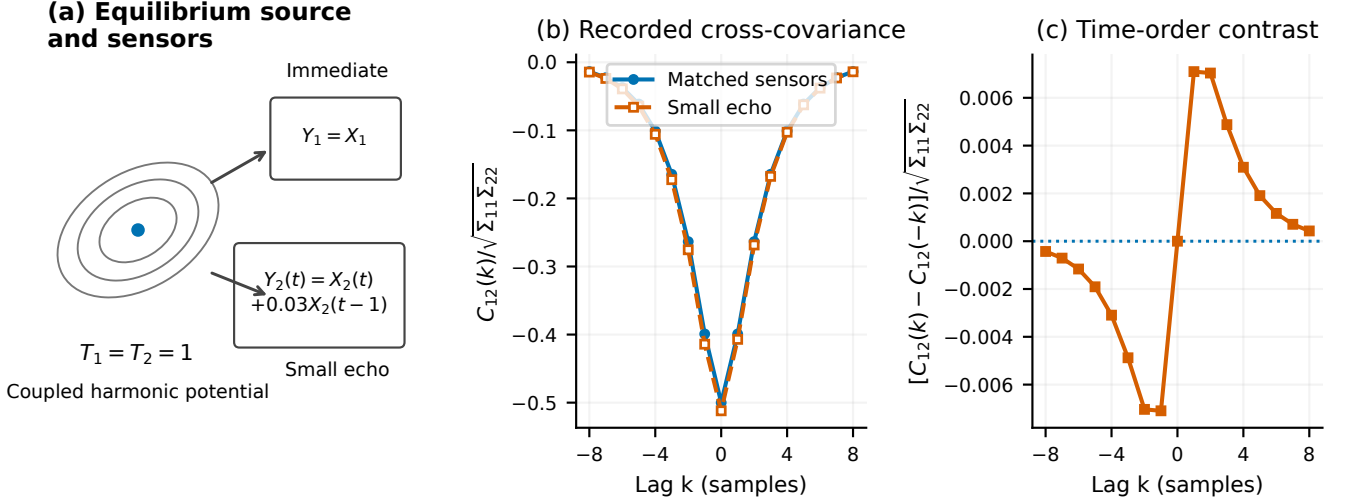


FIG. 1. An equilibrium source can produce an asymmetric recording. Cross-covariance measures how two signals vary together at a time separation, or lag. The Brownian model has a time-symmetric cross-covariance with matched responses. Adding a 0.03 one-sample echo to channel 2 makes the cross-covariance asymmetric while leaving the source, sampling interval, and marginal temperatures unchanged. The odd part is half the difference between the covariance at opposite lags. Both curves use the same source-variance normalization and are exact model calculations, not sample estimates.

covariance bounds are part of the information supplied to the test, not consequences of phase calibration or cancellation.

The physical examples show both the value and the cost of these bounds. In the Brownian model, the calibrated test permits the measured equilibrium asymmetry and detects sufficiently strong driving. Its conservative covariance bound also prevents detection of a weaker driven source at every tested length. We examine this recording cost with known source spectra and with persistence estimated from a reference channel. Finally, we return to the distinction between a sampled trajectory and the underlying motion. A rotating Gaussian source can execute additional full turns between observations without changing its sampled law. Even successful detector correction cannot resolve the resulting ambiguity in entropy production.

II. THE SOURCE AND ITS OBSERVATION CHANNELS

For the equilibrium example, the relevant symmetry concerns the particle before its signals pass through the sensors. Covariance measures how two fluctuations vary together, including their scale. Let $C_{ij}^X(k)$ be the covariance of source channels i and j at a time separation, or lag, k . Because means and covariances determine the Gaussian law, their symmetry can establish reversibility. After subtraction of the mean, a stationary Gaussian process whose coordinates have a common time-reversal parity satisfies

$$\text{reversibility} \iff C_{ij}^X(k) = C_{ij}^X(-k).$$

The equality must hold for every channel pair and lag. It makes the full Gaussian law invariant under reversal, since the mean and covariance determine that law. Equivalently, the source cross-spectral measure is real. The common-parity condition includes position coordinates, which keep their signs under reversal, and separately velocity coordinates, which change their signs. A mixture of positions and velocities requires parity signs in the covariance criterion and is outside the convention used here.

The spectrum expresses the same covariance relations by frequency, where a detector timing change becomes a phase factor. A linear sensor records a weighted sum of source values. This operation changes both the amplitude and timing of the signal. Write its transfer function as $H_i(\omega)$ for channel i and frequency ω . A pure delay has response $e^{-id_i\omega}$, while the echo in Fig. 1 has response $1 + 0.03e^{-i\omega}$. If μ is the matrix spectral measure of the source, the recorded spectral measure is

$$F = D_H \mu D_H^* + F_E, \quad D_H = \text{diag}(H_i). \quad (1)$$

The star denotes conjugate transpose. The source and errors are jointly Gaussian and stationary. Errors are uncorrelated with the source and between channels at every lag, so their spectral measure F_E is diagonal. Unknown constant means are allowed. For $i \neq j$, Eq. (1) multiplies the source cross spectrum by $H_i \bar{H}_j$. Its phase can therefore change even when the source spectrum is real.

We use $C_{ij}(k) = \text{Cov}\{Y_i(t+k), Y_j(t)\} = \int e^{ik\omega} F_{ij}(d\omega)$ for the recorded covariance. Discrete-time densities use $d\omega/(2\pi)$ on $[-\pi, \pi]$. Spectral measures also allow contributions concentrated at individual frequencies. Equiva-

tion (1) describes filters acting on the sampled source sequences. Filtering in continuous time before sampling does not always give this factorization. Distinct continuous frequencies can become one sampled frequency while retaining different detector phases. This aliasing will matter both for the ambiguity result and for the interpretation of phase cancellation.

A. Equilibrium and driving in a trapped particle

We use a Brownian gyrator to connect the statistical symmetry to a physical source [3, 15, 16]. A particle moves in a two-dimensional harmonic trap. The restoring force couples its coordinates, and each coordinate receives random forcing from a heat bath. In the overdamped limit, where inertia is neglected, the dimensionless dynamics are

$$dX = -AX dt + \sqrt{2D} dW_t, \quad (2)$$

$$A = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}, \quad D = \text{diag}(T_1, T_2).$$

Here dW_t supplies the random forcing, and the stationary covariance satisfies $A\Sigma + \Sigma A^\top = 2D$. Equal bath temperatures give equilibrium. Unequal temperatures produce a circulating stationary probability current: trajectories favor one direction around the trap although the position distribution stays constant. We sample at interval one and then apply $H_1 = 1$, $H_2 = 1 + 0.03e^{-i\omega}$. The equilibrium curve in Fig. 1 therefore isolates the effect of unequal detectors within Eq. (1). The same source model with unequal temperatures will test whether detector information permits detection of physical driving.

Throughout the paper, the null hypothesis is a reversible source observed through the allowed channels. A test must control the probability of rejecting this explanation whenever it is true. Failure to reject does not establish reversibility. We also distinguish nonrejection from abstention: a test abstains when a required bound or numerical comparison is unavailable. A source outside the null class is an alternative. Its rejection probability is the detection probability, or power. All records, including abstentions, remain in the denominators of the reported rejection fractions. Table III in Methods collects the main symbols.

III. WHAT A FINITE RECORDING CANNOT DECIDE

The equilibrium example suggests a possible remedy: collect enough data to resolve the source and detector contributions separately. Without restrictions on the source or detectors, this remedy fails. A finite record contains only finitely many covariance lags. A delayed reversible source can agree with an asymmetric source at every retained lag and differ only outside the record.

Consider two explanations for the same two-time record. In the first, the source is asymmetric and the sensors add no delay. In the second, the source is reversible and one sensor delays it. Each channel separately has variance one and no covariance between distinct times, called a unit white marginal spectrum. The first explanation has cross spectrum

$$F_{12}^A = \frac{1}{10} + \frac{1}{5}e^{-i\omega}.$$

A reversible source with $S_{12}^R = \frac{1}{5} + \frac{1}{5}\cos\omega$, observed after delaying channel 1 by one sample, instead gives

$$F_{12}^R = e^{-i\omega} S_{12}^R = \frac{1}{10} + \frac{1}{5}e^{-i\omega} + \frac{1}{10}e^{-2i\omega}.$$

The spectral eigenvalues are at least 7/10 and 3/5, respectively, so both constructions are positive and define valid Gaussian processes. Their cross-covariances are

k	-2	-1	0	1	2
$C_{12}^A(k)$	0	0	1/10	1/5	0
$C_{12}^R(k)$	0	0	1/10	1/5	1/10

A two-time record sees only lags $-1, 0, 1$, the middle three columns. Its complete covariance, and hence its Gaussian law, agrees under both explanations. The extra term in the second spectrum changes only the last column shown. Lag two would distinguish the explanations, but is absent from that record.

The following result extends that construction from a two-time example to every finite stationary Gaussian law with positive-definite covariance. Positive definiteness excludes an exact linear constraint between recorded fluctuations. The constructed sources have finite dependence: their covariances vanish beyond a finite lag.

Theorem 1 (Finite-law ambiguity). *Fix a positive record length and a nonempty finite collection of stationary Gaussian record laws with positive-definite covariance matrices and the same number of channels. One vector of nonnegative integer channel delays gives a reversible Gaussian source representation for every law in the collection. Each source has a strictly positive matrix trigonometric polynomial spectral density and finite dependence. The sources preserve the channel means and variances. No observation error is needed.*

Thus even the exact finite joint distribution cannot exclude a reversible source in this unrestricted class. A test with false-rejection probability at most α throughout that class has detection probability at most α against every nonsingular finite stationary Gaussian alternative. The proof constructs a positive reversible spectrum, then corrects the retained moments exactly (Supplemental Material, abbreviated SM, Sec. S1, Theorem S1 and Proposition S1). An approximate covariance match would not suffice.

For two channels and $N \geq 2$, generic retained covariances require integer relative delay at least $N - 1$. The exceptional records satisfy $C_{12}(\Delta - 1) = C_{12}(\Delta + 1)$ for

an integer $|\Delta| \leq N - 2$. The minimum scale is attainable under an additional covariance condition (SM Sec. S1.2). The source and delays can depend on record length; the result concerns a fixed finite law, not one representation valid at all lengths.

Large delays are not the only source of ambiguity. Suitable prescribed real delays, whose nonreference relative delays are rationally independent with the sampling interval, can be arbitrarily small when continuous frequencies are unrestricted. This means that no nonzero integer combination of those delays and the interval is zero. Aliased frequencies then supply the required channel phases (SM Sec. S2, Theorem S3 and Corollary S2). A spectral envelope instead bounds the effect of small skew (Proposition S3). Bandwidth control therefore matters as well as timing accuracy [17–19]. The next two sections restrict the ambiguity through response calibration or phase cancellation, together with quantitative covariance bounds.

IV. TWO CHANNELS WITH RESPONSE CALIBRATION

The ambiguity result shows why extra information is needed. We first use calibration to bound the relative response phase of a pair of channels. The bound need not remove all detector asymmetry. It must state how much recorded asymmetry remains compatible with a reversible source.

Let v_1, v_2 be the channel variances, which measure the size of their fluctuations. For aligned recordings, use the lag-one contrast

$$\kappa = C_{12}(1) - C_{12}(-1). \quad (3)$$

The contrast compares the same two channels in opposite time orders. It is also the covariance between $Y_1(t) + Y_1(t + 1)$ and $Y_2(t) - Y_2(t + 1)$. Reversal preserves the sum and changes the sign of the difference. A nonzero contrast thus detects one component of recorded asymmetry; other lags can remain asymmetric when this contrast is zero.

Suppose calibration supplies a bound on the relative response phase,

$$r \geq \sup_{\omega} \left| \sin \omega \frac{\text{Im}(H_1 \overline{H}_2)}{|H_1 H_2|} \right|. \quad (4)$$

The ratio is set to zero at a response zero. The supremum covers every allowed response and the source support. Under the reversible-source model,

$$|\kappa| \leq 2r\sqrt{v_1 v_2}. \quad (5)$$

The imaginary part of the cross spectrum changes sign when time order is reversed. The lag-one contrast weights this part by $\sin \omega$. Under the null, that imaginary part can come only from the relative detector phase. The

phase bound and the observed variances therefore limit the detector contribution, as follows. For a reversible source, the cross-spectral measure μ_{12} is real, so

$$\begin{aligned} |\kappa| &= 2 \left| \int \sin \omega \text{Im}(H_1 \overline{H}_2) d\mu_{12} \right| \\ &\leq 2r \int |H_1 H_2| |d\mu_{12}| \\ &\leq 2r\sqrt{v_1 v_2}. \end{aligned}$$

The last step uses positivity of the source spectral measure and Cauchy–Schwarz. Observation noise only increases the marginal variance scale. The factor two is sharp over the spectral-measure class, including line spectra (SM Sec. S3.3). For the Brownian echo response, the exact-model value is $r \simeq 0.0300$, giving a detector allowance near $0.0600\sqrt{v_1 v_2}$. The implemented test uses an upper confidence bound from calibration, not this exact-response value.

A. From a population tolerance to a finite-record test

The population contrast κ is defined by the underlying probability law. An estimate from one finite record fluctuates around this value. To use Eq. (5) for a decision, we must bound those fluctuations without treating correlated observations as independent. Let Γ_N be the covariance of the complete two-channel record, stacked by channel. We require

$$\Gamma_N \preceq q \text{diag}(v_1 I_N, v_2 I_N). \quad (6)$$

This covariance envelope bounds the variance of every weighted sum across both channels and all recorded times. Equivalently, q bounds the largest eigenvalue after each channel is divided by its standard deviation. A valid value must cover the complete dependence structure; response calibration alone cannot provide it.

Physical dynamics can supply this envelope, as in the Brownian example below. Appendix A 1 gives a second construction from source spectral bounds and calibrated response coefficients.

The envelope controls estimation of the contrast and of the variance scale in Eq. (5). For $N \geq 3$, put $n = N - 1$, $U_t = Y_1(t) + Y_1(t + 1)$, and $W_t = Y_2(t) - Y_2(t + 1)$. We center both sequences over the same n starting times, then form their mean product. We also correct each centered sample variance upward to cover the corresponding

population variance:

$$s_n = n^{-1} \sum_{t=0}^{n-1} (U_t - \bar{U})(W_t - \bar{W}), \quad (7)$$

$$V_i = N^{-1} \sum_{t=0}^{N-1} (Y_i(t) - \bar{Y}_i)^2, \quad (8)$$

$$c_N(q) = 1 - q/N - 2\sqrt{q \log(200)/N}, \quad (9)$$

$$D_i(q) = V_i/c_N(q), \quad D_g(q) = \sqrt{D_1(q)D_2(q)}. \quad (10)$$

The denominator $c_N(q)$ accounts for how far a dependent sample variance can fall below the population variance. The resulting ceilings are used only when $c_N(q) > 0$. On one event of probability at least 0.99, both satisfy $D_i \geq v_i$. Their geometric mean therefore bounds $\sqrt{v_1 v_2}$ and is no greater than the larger individual ceiling. A large but valid covariance envelope can make this denominator nonpositive and leave the test unavailable.

The threshold adds an allowance for detector-induced contrast to an allowance for sampling fluctuations. The variance ceilings set the scale of both terms. To bound the fluctuations, we write the centered contrast as a Gaussian quadratic form, a weighted sum of products of Gaussian observations. Its coefficient matrix has Frobenius norm f_n , given below. Two bounds on the covariance-weighted norm are available, so the radius uses their smaller value:

$$f_n^2 = \frac{2n + 2 - 4/n - 4/n^2}{2n^2}, \quad (11)$$

$$R(q, D, N, t) = 2D \min\{q f_n, \sqrt{2qN}/n\} \sqrt{t} + 2qDt/n. \quad (12)$$

Theorem 2 (Calibrated test). *Assume the stationary Gaussian observation model and a simultaneous calibration event of probability at least 0.99 on which $\hat{r} \geq r$ and $\hat{q} \geq q$. The rule rejects when $c_N(\hat{q}) > 0$ and*

$$|s_n| > D_g(\hat{q}) 2\hat{r} + R(\hat{q}, D_g(\hat{q}), N, \log(2/0.03)). \quad (13)$$

Its rejection probability under the stated reversible-source model is at most 0.05. The error allocations are 0.01 for calibration, 0.01 for variance coverage, and 0.03 for the statistic.

The threshold has two terms with different roles. The first permits all lag contrast allowed by the detector phase bound. The second permits sampling fluctuations under the full covariance envelope. Increasing the record length reduces the sampling term, but cannot remove the detector allowance.

Two facts make this population argument survive estimation. Common centering multiplies the contrast's spectral integrand by an even nonnegative factor, so it preserves the detector tolerance. Gaussian quadratic-form bounds then control the centered statistic. On the calibration and variance-coverage events, the fitted threshold is at least a fixed valid threshold. Thus fitting

requires no additional independence assumption. Theorem 2 bounds false rejection over repeated experiments; it does not assign a probability of reversibility to one source after a result. SM Sec. S4.5, Theorem S7 and Lemmas S2–S3 give the proof.

The calibration design must identify the response coefficients and remain valid during testing. The construction uses known deterministic inputs, Gaussian calibration errors, and coefficient regions that cover all frequencies. Appendix A 2 states these conditions and the error allocations. Bounded rounding can be included by enlarging the variance ceilings and contrast allowance (SM Sec. S4.5, Proposition S7).

B. Calibrated inference for the Brownian gyrator

The Brownian model tests both requirements of source inference: allowing an equilibrium detector effect and detecting a driven source. For the family $1 \leq T_i \leq 4$, the drift in Eq. (2) has smallest eigenvalue $1/2$. Its covariance obeys $(2/3)I \preceq \Sigma \preceq 8I$, and $\|e^{-Ak}\| \leq e^{-k/2}$. Together with the response filters, these family bounds give $q = 56$ for every tested temperature (SM Sec. S7.2). This envelope is deliberately common to equilibrium and driven records.

Figure 2 places the different targets side by side. The calibrated test gives no equilibrium rejection at any length. The raw block bootstrap estimates uncertainty by resampling contiguous blocks of the centered lag-product sequence, which retain some temporal dependence. This raw procedure tests recorded asymmetry and has no finite source-null guarantee here. Its detections rise from 87/400 through 271/400 to 397/400. They detect the recorded asymmetry permitted by the equilibrium source and its unequal sensors. Interpreting these raw detections as source irreversibility would therefore become increasingly misleading as data accumulate.

The same valid envelope has a substantial sensitivity cost. The calibrated test detects all 200 fourfold-temperature records only at $N = 524289$, and detects none at temperature ratio two throughout the tested range. The generating entropy-production rates are $9/32$ and $1/16$, respectively, in units of k_B per unit time. Here k_B is Boltzmann's constant. These rates describe the simulated source; the test does not estimate them. A relaxation time measures how long a fluctuation takes to decay by a factor of e . The slowest relaxation time is only two sampling intervals, so the longest record exceeds 260000 relaxation times. Thus this example is limited by a conservative family envelope as well as by correlation.

An exact-response population calculation explains the scale of this cost without changing any study decision. At $N = 524289$ and $q = 56$, the normalized detector and sampling allowances are approximately 0.0600 and 0.0608. Correcting the variance scale gives a threshold near 0.1268. The recorded normalized contrasts at temperature ratios two and four are approximately 0.0915

and 0.1849. These analytic reference values omit calibration and recording increments; they explain why a valid family bound can obscure the weaker alternative. The saved study retains $q = 56$ throughout.

Table I states the comparison targets before interpreting their rejection fractions. SM Secs. S8.5 and S9.4 give their definitions and the phase-boundary controls.

V. THREE CHANNELS WITHOUT PHASE CALIBRATION

Calibration answers the two-channel problem by limiting each allowed detector explanation. A third channel permits a different operation: the detector phases can cancel. In the product of cross spectra for pairs 12, 23, and 31, each channel response appears once and its complex conjugate appears once. Radio interferometry uses this closure-phase cancellation between pairs of antennas [20, 21]. The cancellation is unavailable from two channels alone, since $F_{12}F_{21} = |F_{12}|^2$ is real for every source.

Let $F_{ij}(\omega)$ and $S_{ij}(\omega)$ be the recorded and source cross-spectral densities. These densities refer to the absolutely continuous parts of the spectral measures. Under Eq. (1),

$$F_{12}F_{23}F_{31} = |H_1H_2H_3|^2S_{12}S_{23}S_{31}. \quad (14)$$

The remaining detector factor is real and nonnegative. Consequently, an imaginary part in this product must come from the source. We use

$$J(\omega) = \text{Im}\{F_{12}(\omega)F_{23}(\omega)F_{31}(\omega)\}. \quad (15)$$

A reversible Gaussian source has real cross spectra and hence $J = 0$ almost everywhere. Absolute summability of the observed cross covariances will make these spectra continuous, so the equality holds at each tested frequency. A nonzero J then excludes a reversible source under the sampled response model. The converse fails because source phases can also cancel around a cycle.

For pure delays, the cancellation is explicit: the phase sum is $-\omega[(d_1 - d_2) + (d_2 - d_3) + (d_3 - d_1)] = 0$. Figure 3 compares this operation with the calibrated allowance. It also exposes the next obstacle. The identity applies to population spectra, whereas a finite record supplies estimates over only a limited range of lags. To obtain a test, we must control both estimation error at retained lags and the contribution of unobserved lags. These extra restrictions are why the cycle test can coexist with the finite-law ambiguity.

A. Uncertainty on the three spectral edges

We estimate each cross spectrum from sample covariances up to a fixed lag cutoff L . A larger cutoff retains more of the true spectrum but also introduces

more estimated covariances. Assume absolute summability of the three observed cross-covariance sequences. Fix $0 \leq L < N$ and G frequencies before observation. The required information is a covariance envelope and a bias allowance:

$$\Gamma_N \preceq D_K, \quad (16)$$

$$\sum_{|k| > L} |C_{ij}(k)| + \frac{1}{N} \sum_{|k| \leq L} |k| |C_{ij}(k)| \leq B_{ij}, \quad (17)$$

Here $D_K = \text{diag}(K_i I_N)_{i=1}^3$, with positive K_i . Equation (17) holds for $ij \in \{12, 23, 31\}$, with nonnegative B_{ij} . The first term of each bias allowance covers lags beyond the cutoff; the second covers finite lag weights. As in the calibrated test, the covariance envelope covers the full record. Separate bounds on the three marginal covariances do not imply it.

Set $P_N = I - \mathbf{1}\mathbf{1}^\top/N$, $h = 2L + 1$, and

$$[W_\omega]_{st} = \mathbf{1}\{|s - t| \leq L\} e^{-i\omega(s-t)},$$

$$\hat{F}_{ij} = N^{-1}(P_N y_i)^\top W_\omega (P_N y_j).$$

The three channel pairs form the edges of the spectral triangle. For error probability δ_e , put $t_\delta = \log(12G/\delta_e)$. A simultaneous error radius for each edge is

$$e_{ij,\delta} = B_{ij} + \sqrt{K_i K_j} \left(\frac{3h}{N} + 2\sqrt{\frac{ht_\delta}{N}} + \frac{\sqrt{2ht_\delta}}{N} \right). \quad (18)$$

The allowance B_{ij} covers the covariance truncation and finite lag weights. The $3h/N$ term covers centering, and the remaining terms cover sampling fluctuations. Increasing the cutoff retains more covariances and reduces the omitted tail, but it also increases h in the sampling terms. The fixed cutoff must therefore be assessed through the complete radius. These edge errors must be propagated through the product before a nonzero estimated cycle can support rejection. For $x = (|\hat{F}_{12}|, |\hat{F}_{23}|, |\hat{F}_{31}|)$ and the corresponding $e = (e_{12}, e_{23}, e_{31})$, define

$$\mathcal{R}_\times(x, e) = \prod_{j=1}^3 (x_j + e_j) - \prod_{j=1}^3 x_j. \quad (19)$$

Expanding this expression gives all seven product terms that contain at least one edge error. The logarithm in the edge radii covers six real components, two tails, and G frequencies at total error δ_e . Thus every tested cycle is compared with the uncertainty in its full product.

Theorem 3 (Cycle test). *Under the sampled stationary Gaussian reversible-source model and Eqs. (16)–(17), reject if at least one of the G fixed frequencies satisfies*

$$|\text{Im}(\hat{F}_{12}\hat{F}_{23}\hat{F}_{31})| > \mathcal{R}_\times(x, e_\alpha). \quad (20)$$

The rejection probability is at most α . Equality does not reject.

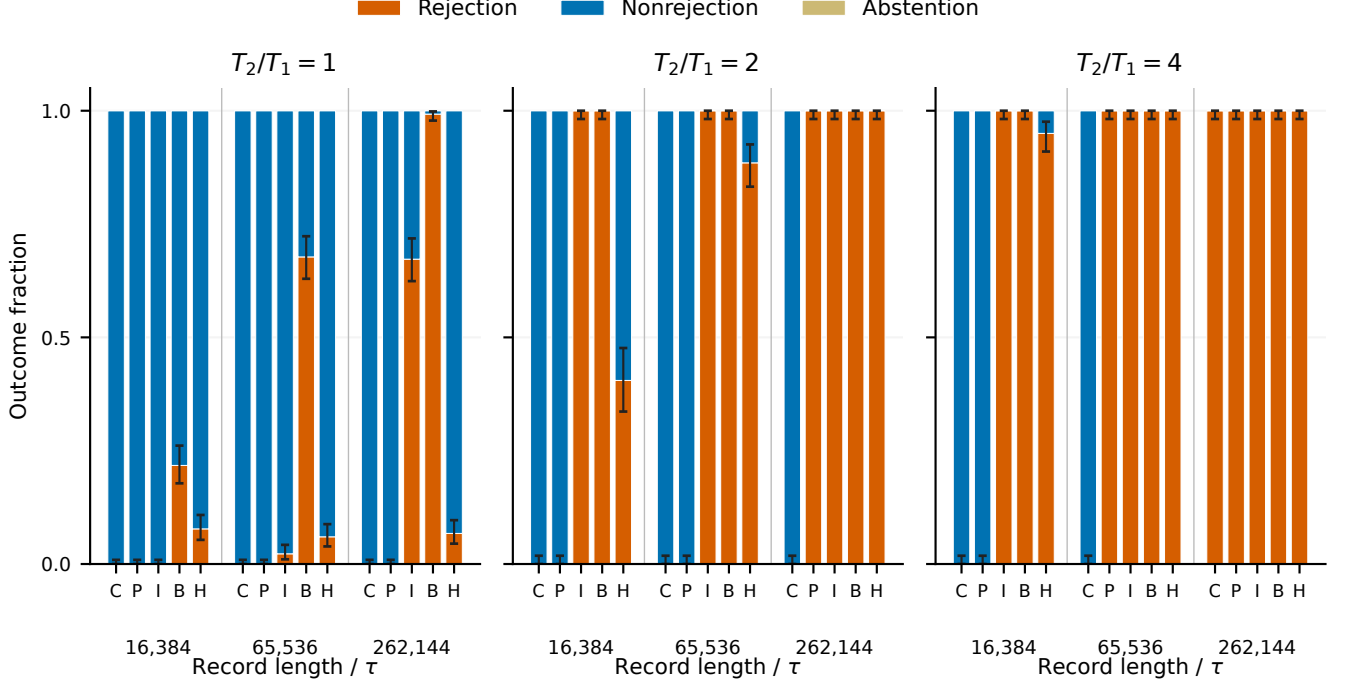


FIG. 2. Raw bootstrap detections reach 397/400 at equilibrium, while the calibrated source test rejects 0/400. Temperature ratios are 1, 2, and 4; lengths are $N/2$ relaxation times. Methods: C, calibrated; P, phase ignored; I, independent windows; B, moving-block bootstrap; H, imaginary coherency. Equilibrium cells have 400 records; alternatives have 200. Bars give pointwise 95% binomial intervals. Raw methods test recorded asymmetry; C tests source reversibility under response information. No method abstains. Values are saved study results.

TABLE I. Comparison targets. An approximate test for recorded asymmetry is not a finite-sample test for source reversibility under unequal sensors.

Procedure	Null restriction	Detector information	Guarantee here
Calibrated contrast	Reversible source within the allowed response model	Relative phase and full covariance bound	Finite false-rejection bound
Spectral cycle	Real source cross spectra under sampled diagonal responses	Factorization; covariance and tail bounds	Finite false-rejection bound
Independent-window lag rule	Zero recorded lag contrast	None	Ignores overlapping-window dependence
Moving-block bootstrap	Zero recorded lag contrast	None in raw form	Approximate bootstrap
Imaginary coherency	Zero imaginary normalized cross spectrum	None	Approximate segment bootstrap
Phase-adjusted comparators	Recorded contrast within a detector allowance	Same phase bound as calibrated rule	No finite source-null guarantee here

The rule rejects when edge estimation errors cannot explain the imaginary part of the estimated product. Its proof normalizes channel i by $\sqrt{K_i}$, applies Gaussian quadratic-form bounds to each pairwise spectrum, and propagates those bounds through the product. This accounts for overlapping samples and the fixed frequency set (SM Sec. S4.2, Theorem S5). Bounded recording errors enlarge the edge radii as in SM Sec. S4.3.

B. A rotating source and the cost of the edge bounds

The cycle identity removes detector phase exactly, but useful detection still depends on the size of the edge errors. To examine that cost, we extend the trapped-particle example to three coordinates. A linear force restores the particle toward the center while a rotational

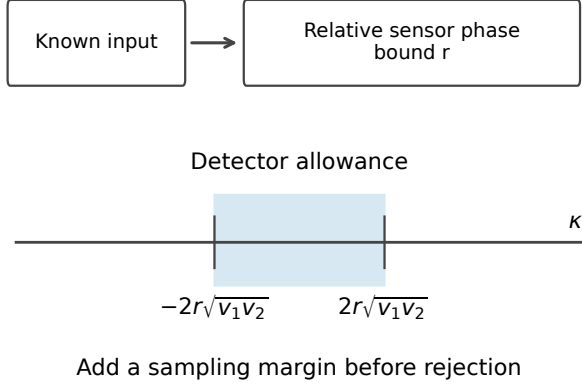
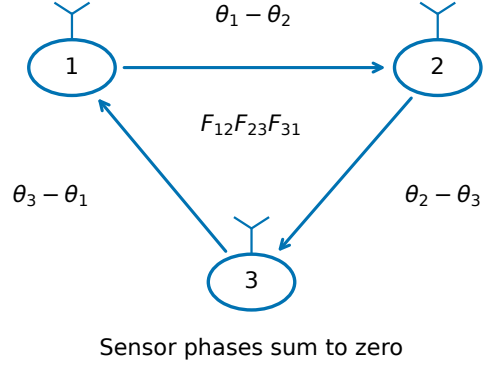
(a) Two channels: calibrate**(b) Three channels: cancel sensor phase**

FIG. 3. Calibration bounds detector phase, while a three-channel closure product cancels it. The calibrated route allows a nonzero source-null lag contrast and adds a sampling margin. The cycle route uses the three oriented cross spectra; each channel phase appears once with each sign. Both routes still require a full covariance bound. The spectral route also needs a bound on omitted lags.

force circulates it about the $(1, 1, 1)$ axis. The model has

$$A = \lambda I + \Omega, \quad D = \lambda \Sigma_\xi, \quad \Omega = \nu \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}, \quad (21)$$

where $\Sigma_\xi = (1 - \xi)I + \xi \mathbf{1}\mathbf{1}^\top$. For $\lambda = \log 4$ and $\nu = 2\pi/(3\sqrt{3})$, one sampling interval gives a $2\pi/3$ rotation and a cyclic coordinate permutation, attenuated by $1/4$. The equilibrium control sets $\nu = 0$. Both use $\xi = 1/4$ and the same family bounds.

Over $0 \leq \xi \leq 1/4$ and $\lambda \geq \log 4$, the source spectral ceiling is $5/2$. If g_i is the absolute sum of detector coefficients, valid diagonal ceilings are $K_i = (5/2)g_i^2$. Geometric covariance decay also gives the tail allowances at $L = 8$ (SM Sec. S7.3). The filters have principal delays zero, one, and two samples, with small later echoes. These bounds apply to the complete covariance; they are not three independent marginal assertions.

Figure 4 shows when these allowances permit rejection. The single-frequency rules test $\pi/2$; the two-frequency rule tests $\{\pi/4, \pi/2\}$ and uses the larger error allowance required for both frequencies. At $N = 1048576$, the common and diagonal single-frequency rules each reject $1/200$ rotating records; the two-frequency rule rejects $200/200$. All three reach $200/200$ at $N = 2097152$, while equilibrium controls give no rejections. The gain comparison in SM Fig. S3 explains why diagonal bounds matter: they preserve all $200/200$ rejections across the six transported gains, while a common scalar ceiling loses them at four gains. Valid covariance inflation can also remove detection (SM Fig. S4).

C. What the cycle preserves and what it cannot identify

The separate edge radii allow the rule to respect different channel scales. The rule is unchanged by an invertible change of channel units when every bound undergoes the corresponding transformation. Under $z_i \mapsto a_i z_i + b_i$, $a_i \neq 0$, use $K_i \mapsto a_i^2 K_i$, $B_{ij} \mapsto |a_i a_j| B_{ij}$, and $\eta_i \mapsto |a_i| \eta_i$. The statistic and threshold both multiply by $(a_1 a_2 a_3)^2$. This invariance changes data and all bounds together. Reducing signal gain while keeping added noise fixed changes the signal-to-noise ratio and is a different experiment. Response zeros can erase the witness, and numerical limits can still prevent a decision.

At one frequency, channel phases can make all cross spectra real exactly when every product around a cycle of nonzero edges is real. One proves this by first making the edges of a spanning tree positive. A spanning tree connects all channels without a cycle. Each remaining edge closes a cycle. For a Markov chain, the Kolmogorov criterion equates the products of forward and reverse transition rates around every loop [22]. Here the products contain complex cross spectra rather than non-negative rates. Spectral products can be negative real, and the condition does not construct causal responses across frequencies. A real cycle is compatible with reversibility but also with sources whose asymmetry this frequency does not reveal.

The full covariance ceiling remains necessary for this finite-window argument even when it is conservative. Replacing it by only the spectral value at the tested frequency is invalid under these assumptions. For independent channels $Y_i(t) = Z_i(t) + Z_i(t - 2)$, each spectrum vanishes at $\pi/2$, but a finite cross-spectral estimate has positive variance there. A finite window mixes nearby fre-

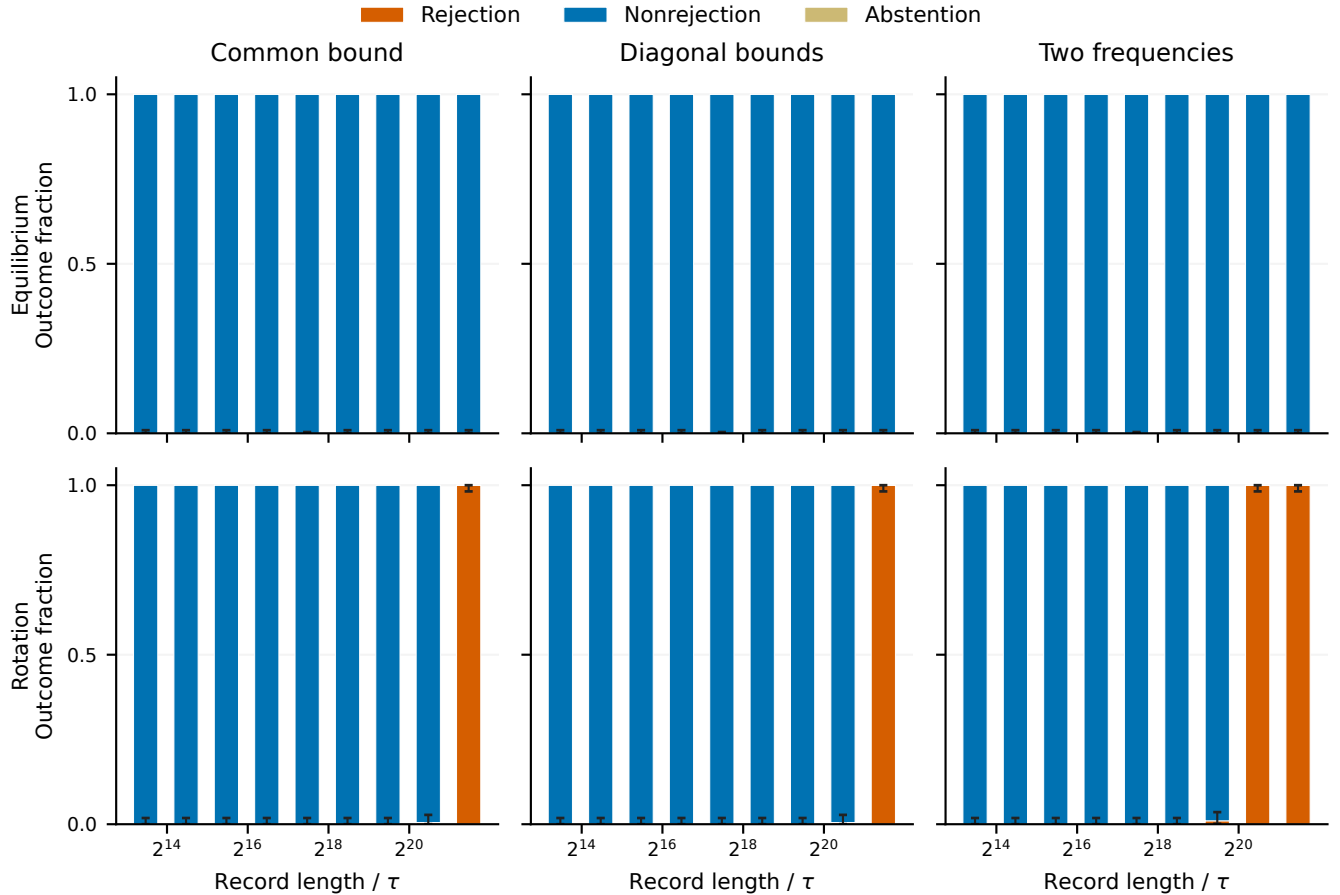


FIG. 4. The two-frequency rule detects rotation at a shorter tested length than either single-frequency rule. Lengths are shown as $N \log 4$ decay times. Fractions use all 200 rotating records per length; equilibrium cells have 400, except 1000 at $N = 131072$. Bars are pointwise 95% binomial intervals. The two-frequency error allowance is included. Common and diagonal decisions coincide in this primary comparison. All methods have zero abstentions. These empirical fractions do not replace the sufficient power condition.

quencies. SM Secs. S4.2–S4.3 give this counterexample, the gain result, and a sufficient power condition.

VI. HOW MUCH DATA DOES INFERENCE REQUIRE?

The Brownian and rotational examples required long records before their valid thresholds permitted detection. How much of this cost is unavoidable, and how much comes from conservative bounds? We first set aside calibration and source estimation. Even with exact responses and known marginal spectra, a directional signal can occupy a narrow frequency range that a short record resolves poorly.

Consider unit-variance Gaussian channels with identity responses and marginal spectral densities between $1/2$ and $S/2$, where $S \geq 3$. We seek an absolute covariance contrast of at least γ , with $0 < \gamma \leq 1/4$. The difficult alternative used below places its contrast in two narrow

bands. The record covariance can be decomposed into modes, each a weighted combination of observed values. Their total overlap with the two bands is $N/(\mathcal{S}-1)$, which measures the effective number of modes that probe the directional signal. Narrow bands reduce this number, so many observations can still provide little evidence against a reversible source.

We quantify the distinction through the relative entropy between the two finite-record laws. If this quantity is too small, no test can attain the desired false-rejection and detection probabilities. This use of relative entropy concerns statistical distinguishability. The same measure applied to forward and reversed trajectories has a thermodynamic interpretation under the appropriate physical assumptions [23, 24].

Theorem 4 (Recording cost). *For each $(\mathcal{S}, \gamma)$ in this class, SM Sec. S6 constructs a fixed reversible source and a fixed alternative with contrast $-\gamma$ and the same known marginal spectra. The necessary length inequality*

in SM Sec. S6.3 applies to any test of this pair with false-rejection probability α and detection probability $1 - \beta$, where $0 < \alpha, \beta < 1$ and $\alpha + \beta < 1$. For the full class, the calibrated test with exact identity responses and $q = \mathcal{S}$ has false-rejection probability at most 0.05 and detection probability at least 0.95 whenever

$$N \geq 1024\mathcal{S}/\gamma^2. \quad (22)$$

At $\mathcal{S} = 9$, $\gamma = 0.1$, and $\alpha = \beta = 0.05$, at least 5214 samples are necessary for the fixed pair; direct evaluation of the sharper sufficient condition gives 129580 samples for the full class.

Thus dependence can force thousands of samples even with exact responses. SM Sec. S6, Theorems S8–S9, gives the complete necessary inequality, its finite trace-defect bound, and the sufficient calculation. The information required by the two error rates is $(1 - \beta) \log[(1 - \beta)/\alpha] + \beta \log[\beta/(1 - \alpha)]$. A record is necessarily too short if an upper bound on its available relative entropy is smaller. The sufficient-to-necessary length ratio is below 25 in the worked case. It compares a class-wide sufficient bound with a fixed-pair lower bound, so it is not an optimal-length claim.

A. Unknown persistence: why the whole confidence set matters

The preceding length bounds assume that the source spectra are known. When dependence must be estimated, uncertainty about the source also enters the threshold. We study this additional cost with a noiseless reference channel from a stationary Gaussian first-order autoregression, or AR(1). Persistence describes how strongly a source value remains related to later values. In the AR(1) model, each centered reference value is ϕ times its predecessor plus a new independent Gaussian random input, called an innovation. For $0 < \phi < 1$, correlations decay on the scale $-1/\log \phi$, approximately $1/(1 - \phi)$ samples near one. We therefore use $(N - 1)(1 - \phi)$ to express record length in approximate correlation times. At $\phi = 0.995$, the lengths 16385, 32769, and 524289 span about 82, 164, and 2621 such times.

Known persistence would permit the common filter $(-\phi, 1)$ to remove the autoregressive spectral variation. Applying the same filter to both channels leaves their relative detector phase unchanged. Replacing ϕ by a point estimate, however, does not bound the variation left by estimation error. We instead retain all persistence values accepted by a finite Gaussian test of cumulative prediction errors. The resulting two-sided confidence set contains the true value with probability at least 0.995 over repeated recordings. SM Secs. S5.1–S5.3 give its exact law and conservative numerical cutoffs.

To transfer this set to the pair, both Gaussian source marginals must have the same persistence. The reference response is a nonzero scalar, the second response

is a nonzero finite impulse response, and both channels are noiseless before bounded recording. Their cross spectrum is arbitrary subject to positivity and is real under the null. Independent calibration covers the unchanged second response. These assumptions connect reference information to both testing channels.

Let $[\ell, u]$ be the smallest interval containing all accepted persistence values, called the confidence hull. For $u < 1$, define $\Lambda(x) = (1 + x)/(1 - x)$ and $z(x) = \log \Lambda(x)$. The AR(1) spectral maximum-to-minimum ratio is $\Lambda(\phi)^2$, so this transform measures the spectral variation that the covariance bound must cover. For a fixed common-filter coefficient a ,

$$\mathcal{S}_a = \exp\{z(u) - z(\ell) + 2|z(a) - z_c|\}, \quad z_c = \frac{1}{2}[z(u) + z(\ell)]. \quad (23)$$

The transformed width $z(u) - z(\ell)$ is the cost of source uncertainty. The second term is the cost of choosing a filter away from the transformed center. Equation (23) therefore connects the geometry of the accepted set to the covariance envelope used by the test. Coverage alone is insufficient: a set can contain the truth yet produce an uninformative threshold.

Let α_B be the source-set error, α_L an additional lag-set error, and β the common energy-event failure probability.

Theorem 5 (Source-envelope consequence). *Assume the two-sided source set and common energy event specified in SM Sec. S5.4, under the positive-denominator conditions stated there. On source coverage and that event, the confidence hull is nonempty, $u < 1$, and*

$$\frac{\Lambda(u)}{\Lambda(\ell)} \leq 2C_{\text{gap}}. \quad (24)$$

The constant C_{gap} is explicit in the full statement in SM Sec. S5. The probability is at least $1 - \alpha_B - \beta$, or $1 - \alpha_B - \alpha_L - \beta$ for the lag intersection.

This bound controls the span of every accepted persistence value, even when the confidence set has separate pieces. Excluding persistence one makes the envelope finite, but does not bound how large it can be. The additional energy inequalities provide that control (SM Secs. S5.4–S5.5). At persistence 0.8, a target probability 0.75 of a nonempty set excluding one requires at least 65 differences, while 256 suffice under error allowance 0.005 (SM Sec. S5.7). This factor-of-four bracket concerns end-point exclusion, not rejection.

The test uses a finite list of filters fixed before observation. For M members, the source, calibration, variance, and contrast allocations in Table IV cover the complete union decision. A length-dependent list $a_j = 1 - 2^{-j}$ controls placement near persistence one, whereas a fixed list bounded away from one cannot do so uniformly. SM Sec. S5.8 gives the rule, including empty sets and unavailable members.

B. A complete detection example and its observed limits

A useful source confidence set is still only one requirement for detection. Calibration, variance coverage, the contrast margin, and numerical completion must hold together. The following delayed-source example establishes this joint result at a specified long length. It also gives a direct comparison with the shorter records, where one or more sufficient conditions fail.

a. Example: a delayed persistent source. Let X, Z be independent unit-variance Gaussian AR(1) processes with $\phi = 0.995$. Observe

$$Y_1(t) = 2 + X_t, \quad Y_2(t) = -1 + 0.9X_{t-1} + \sqrt{0.19}Z_t.$$

Use identity responses, $N = 524289$, independent 256-row two-coefficient calibration with noise standard deviation 0.01, and the bank $(0, 0.5, 0.8, 0.9, 0.97, 0.995)$. Values lie on the 2^{-20} grid within 2^{-21} of the ideal samples. Under the complete design and bounded arithmetic specified in SM Secs. S5.9–S5.10,

$$\Pr(\text{reject}) \geq 0.91 - (4N + 512)e^{-128} > 0.90. \quad (25)$$

The full parameter statement is retained in SM Sec. S5. The bound includes completion of all calculations needed for rejection. Earlier points on the doubling grid fail a source, variance, or contrast-margin condition; the long-record result gives no corresponding shorter-record guarantee.

The saved synthetic comparison shows how these requirements affect decisions (Fig. 5). Table II shows the transition from unavailable or inconclusive tests to detection. The lag set abstains throughout the two shorter cells because its upper endpoint still reaches one. SM Fig. S7 locates the first failed condition, and Figs. S1–S2 and S8 separate coverage from endpoint availability.

The improvement is conditional. At $N = 8193$, the standalone set rejects 46/200 and its split-budget intersection rejects 32/200. Their paired difference is 0.070, with conservative 95% interval $[-0.006, 0.142]$. At moderate persistence the intersection can instead retain detections lost by the standalone set. The two-sided threshold diagnostics at $N = 524289$ have median phase allowance about 7.6×10^{-5} and median sampling radius about 5.0×10^{-4} (SM Figs. S5–S6). These saved summaries show that estimating dependence remains costly even when phase calibration is accurate.

VII. WHAT SAMPLING HIDES

The preceding tests attribute asymmetry to the sampled source. To relate that conclusion to thermodynamics, one must also account for motion between observations. The rotating model makes this remaining ambiguity explicit. Adding a full turn between samples changes

TABLE II. Two-sided source inference at $\phi = 0.995$ and strong delay. Each row uses 200 independent records. R, NR, and A denote rejection, nonrejection, and abstention.

N	$(N - 1)(1 - \phi)$	R	NR	A
16385	81.92	153	29	18
32769	163.84	196	3	1
524289	2621.44	200	0	0

the continuous motion without changing the observed transition.

Set $\nu = 2\pi/\sqrt{3}$ in the model of Sec. V. Rotation then completes a full turn per sample, and the transition is $I/4$. The sampled Gaussian source is reversible although its continuous-time probability current is nonzero. The detector-phase cancellation remains valid, but it cannot restore time order absent from the source samples.

For this reason, the entropy-production rate requires the continuous-time dynamics, not only their sampled law. For overdamped coordinates with even reversal parity, the stationary rate is [24]

$$\sigma = \text{tr}\{(A - D\Sigma^{-1})^T D^{-1}(A - D\Sigma^{-1})\Sigma\}. \quad (26)$$

It measures irreversibility in units of k_B per unit time and vanishes at detailed balance for these models. The rotational rate is $6\nu^2/\lambda$. More generally,

$$\nu_k = (2\pi/3 + 2\pi k)/\sqrt{3}, \quad k \in \mathbb{Z}, \quad (27)$$

gives the same sampled transition and innovation covariance for every integer k , while $6\nu_k^2/\lambda$ grows without bound. SM Sec. S7.4 derives both the aliasing identity and the entropy rates. Thus even complete knowledge of the sampled law cannot determine continuous-time entropy production without additional dynamical or bandwidth information.

VIII. DISCUSSION

An asymmetric recording becomes evidence against a reversible source only after the allowed observation mechanisms have been constrained. The finite-law construction shows why this information is necessary. Calibration and spectral cycles then provide two distinct population constraints: one bounds the phase that the detectors can add, and the other cancels it. Neither constraint by itself supplies a finite-record decision. The complete covariance and, for the cycle, the omitted-lag bound determine whether the observed departure can exceed sampling uncertainty.

The Brownian example gives this distinction a physical interpretation. Its equilibrium records become easier to distinguish from a reversible *recorded* process as length increases. The raw bootstrap detections are therefore expected under unequal sensors. The calibrated test asks

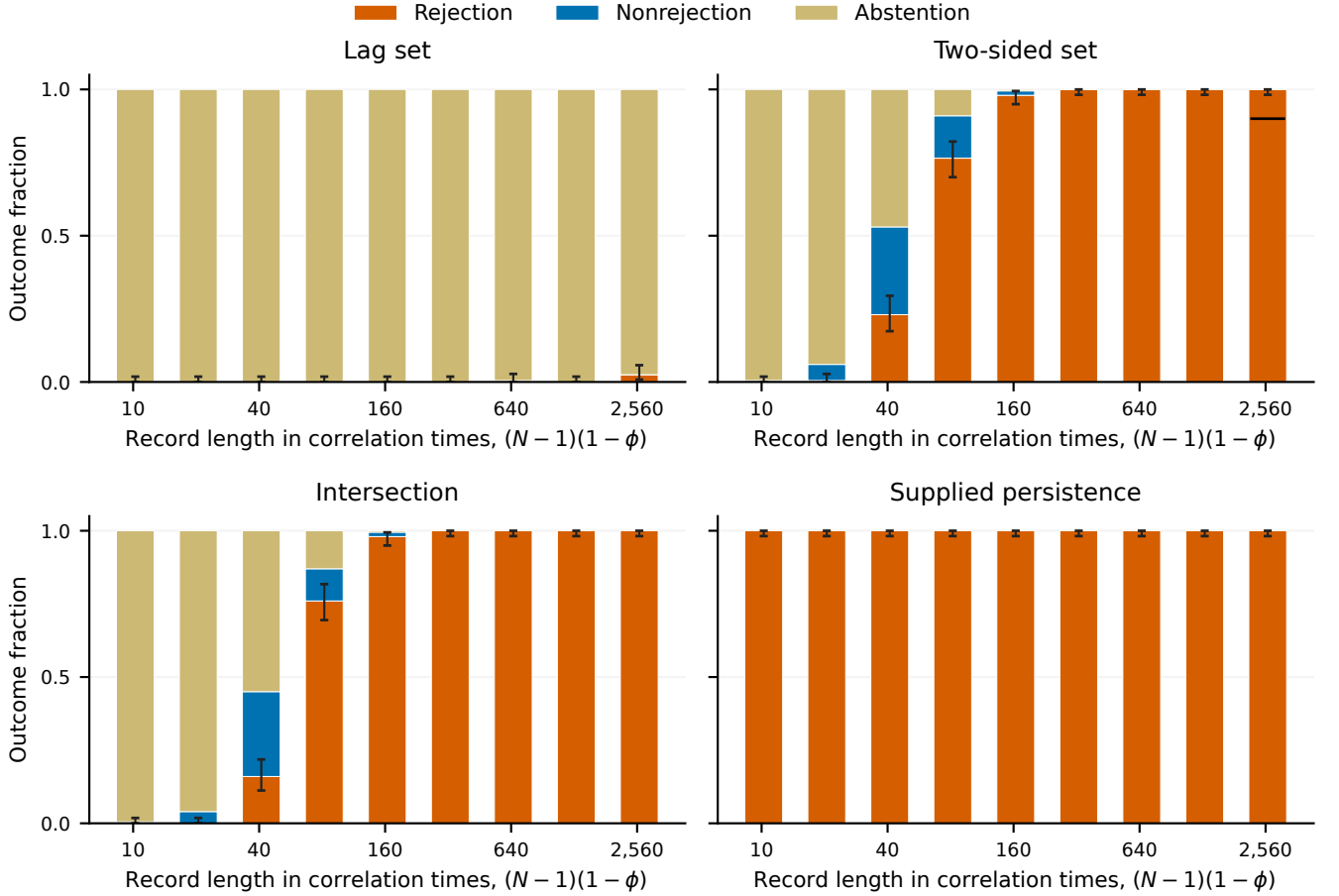


FIG. 5. Estimating persistence delays detection even when the supplied-persistence test rejects every alternative record. Outcomes are shown against approximate correlation times $(N-1)(1-\phi)$ at $\phi = 0.995$, with 200 independent records per length. Lag confidence often reaches persistence one and therefore causes abstention. Two-sided confidence becomes useful sooner; intersection is not uniformly better. Rejection intervals are pointwise 95% binomial intervals. The ideal-model lower bound at the longest length is distinct from the observed fraction.

a different question and retains the equilibrium explanation. Its failure to detect the twofold-temperature alternative at every tested length also matters. A valid source test can be too conservative to reveal actual driving. The recording-cost result establishes a dependence cost even with exact information, while the persistence examples show the further cost of learning a usable covariance envelope.

These results connect established operations to the same source question. Closure products remove local phase [21]; Gaussian quadratic-form inequalities control dependent estimates [25–27]; and autoregressive inversion gives confidence sets [28–30]. Their combination must allow a nonzero detector contribution and cover the whole accepted source set. The two-sided construction improves several persistent conditions, but intersecting it with lag confidence changes detection in both directions because the error allocations change. A more informative confidence set does not automatically yield a more powerful final decision.

Comparisons must also preserve the object being tested. Bi and Calhoun’s coherence witness concerns shared hidden input [31]. Their multiplicative response model filters independent noise and shared input together, so response factors cancel under an independent-probe null. Our null permits correlated reversible sources. Raw lag and imaginary-coherency procedures concern recorded asymmetry instead. The near-boundary controls illustrate the consequence: raw procedures reject all 400 records, while calibrated procedures reject none, even when detector-induced contrast reaches 99% of its population tolerance (SM Sec. S9.4). These outcomes answer different questions about the same records.

For experimental use, the first requirement is a justified observation model. The calibrated route needs a bound on relative response phase. The cycle route needs sampled diagonal factorization and errors uncorrelated between channels. Both need a full covariance bound from physical dynamics or a valid confidence procedure, and the cycle route also needs the covariance tail. These

quantities determine a threshold that can be assessed before a nonzero statistic is given a source interpretation. The examples show why a record measured in many correlation times can still be too short when the available bounds are broad.

All guarantees here assume stationarity, joint Gaussian laws, stable linear responses, fixed channel identities and common reversal parity, and the stated covariance and error bounds. Source fitting further requires a noiseless scalar reference and shared autoregressive marginal persistence. Reference noise, non-Gaussian innovations, correlated detector errors, and underestimated bounds fall outside the corresponding guarantees. The sensitivity analyses cover only their tested conditions, and the physical examples are synthetic. Bounded recording corrections cover a valid rounding error; they do not make a finite random generator exactly Gaussian.

Finally, source reversibility at the sampling interval is distinct from continuous-time dissipation. Relative entropy between forward and reversed trajectories constrains dissipation under the required observation and dynamical assumptions [2, 24, 32]. The aliased rotations show that identical sampled laws can nevertheless correspond to arbitrarily different entropy-production rates. Detector information resolves one source of ambiguity, while bandwidth or dynamical information is needed for the other. Within the stated model, the tests identify when detector effects and finite-record uncertainty no longer suffice to explain the observed time asymmetry.

Appendix A: Methods

The observation filters, alignment anchors, tested contrasts, frequency sets, model families, sample counts, and analysis rules were fixed before the records for the present study were generated. No method or parameter was selected from the outcomes of the present study. The study has 69 conditions and 20800 independent records. Paired transformations and numerical settings do not add independent records.

The source pair is $X_1(t) = X(t)$ and $X_2(t) = cX(t - d) + \sqrt{1 - c^2}Z(t)$, where X, Z are independent stationary unit-variance Gaussian AR(1) processes. The null has $(d, c) = (0, 0.9)$; the alternatives have $(1, 0.4)$ and $(1, 0.9)$. Recording means are 2, -1 . The principal detector responses are identities. A separate 256-row calibration fits two coefficients using orthogonal sign columns and independent mean-zero Gaussian errors of standard deviation 0.01. Both coefficients are fitted when one true coefficient is zero. The primary fixed bank is $(0, 0.5, 0.8, 0.9, 0.97, 0.995)$. A separately declared bank uses $1 - 2^{-j}$ through $j = \lceil \log_2(N - 1) \rceil$.

The calculations use exact recorded moments and arithmetic bounds that enclose each result conservatively. Unresolved comparisons cause abstention; unexpected execution failures remain separate. SM Sec. S8.6 gives the precision settings and work limits.

TABLE III. Main notation. Source and recorded covariances are distinguished by the superscript X .

Symbol	Meaning
$N, n = N - 1$	Samples and adjacent pairs.
F_{ij}, S_{ij}	Recorded and source spectral densities.
κ, r	Lag contrast and relative-phase bound.
q, K_i	Full-record covariance bounds, standardized or channel scaled.
L, B_{ij}	Retained lag cutoff and omitted-lag bias allowance.
ϕ, ψ, a_j	True persistence, candidate persistence, common-filter coefficient.
$S_a, \Lambda(x)$	Residual spectral ratio and $(1+x)/(1-x)$.

The principal recordings have 20 fractional bits and include the corresponding bounded-error correction. Paired grids with 8 and 12 fractional bits isolate rounding. Separate finite 8-bit and 12-bit recorders span $[-16, 16]$, with clipping to the outer bin centers. Saturation without a valid perturbation certificate causes abstention. Calibration stays at the primary recording precision in these comparisons.

The moving-block bootstrap [33] uses 499 resamples and block length $\max\{2, \lceil n_b^{1/3} \rceil\}$ for the centered product sequence. Here $n_b = N - 2$ is the retained product count after omitting row zero, as specified in SM Sec. S8.5. Its probability estimate uses a plus-one correction. Imaginary coherency [34] uses nonoverlapping 256-sample segments at frequency $\pi/2$ and the declared bootstrap of segment contributions. An incomplete final segment is omitted only for this comparator. Missing phase information causes the phase-adjusted comparators to abstain.

Each cell has a pointwise two-sided 95% Clopper-Pearson confidence interval [35]. A paired rejection difference uses the two discordant outcomes. Subtracting simultaneous 97.5% intervals for their probabilities gives a conservative 95% difference interval. Each interval quantifies uncertainty in its own rejection probability or paired difference over repeated studies. These intervals are pairwise, without a familywise guarantee across all comparisons. Rejection among available tests and threshold quantiles are descriptive. Exact rational comparisons determine source-set coverage and whether its upper endpoint is below one before numerical summaries are formed.

The recorded study environment uses Python 3.9.6, NumPy 2.0.2, SciPy 1.13.1, and Matplotlib 3.9.4 [36–38]. Certified probability integrals used Python 3.12.14, NumPy 2.5.3, and python-flint 0.9.0. The package retains the certificates and environment files. Empirical fractions concern the declared finite generator; the mathematical power bound concerns the ideal Gaussian law followed by bounded recording.

TABLE IV. Error allocations explain the logarithms in the thresholds. Each column totals 0.05. The fitted-source bank has M fixed filters.

Component	Calibrated pair	Fitted-source bank
Source confidence	—	0.005
Calibration	0.01	0.01
Both variance ceilings	0.01	0.01
Contrast	0.03	0.025
Variance logarithm	$\log(200)$	$\log(200M)$
Contrast logarithm	$\log(2/0.03)$	$\log(80M)$

1. Constructing the covariance envelope

One construction uses a source spectral floor and ceiling. If normalized source marginal densities obey $0 < m_0 \leq g_i \leq M_0$, define $\mathcal{S} = M_0/m_0$. If calibration bounds $\sup |H_i|^2 / \|h_i\|_2^2$ by $\hat{C}_i$, and the error spectra obey $f_{E,i} \leq J_i \text{Var } E_i$, then

$$\hat{q} = \max\{2\mathcal{S} \max_i \hat{C}_i, \max_i J_i\} \quad (\text{A1})$$

is valid when the calibration confidence regions contain the true response coefficients. Here h_i is the vector of response coefficients. The source ratio and the error bounds are supplied separately. Physical dynamics can also supply the full bound directly, as in Sec. IV B.

2. Response calibration and error allocation

Calibration uses known deterministic inputs and a full-rank response design at fixed integer coefficient positions. Errors are independent Gaussian values with a common positive variance within each channel, and residual degrees of freedom are positive. Responses must stay unchanged between calibration and testing. Coefficient confidence regions and derivative bounds between grid points give an all-frequency phase allowance (SM Secs. S3.4–S3.5). Table IV gives the total false-rejection budget. Bounded rounding is covered by enlarging the variance ceilings and adding a deterministic contrast error (SM Sec. S4.5, Proposition S7).

3. Scientific timing

The supplied-source results were selected for comparison after their outcomes were known. The cycle and cumulative-innovation methods were developed after lag-ratio outcomes were known. Diagnostics of source-envelope geometry and threshold components were selected after inspection of the corresponding four-method comparison. They summarize its saved confidence sets and thresholds without changing decisions. The two-sided method and the present study design were developed with that evidence available, then fixed before their

own fresh records were generated. The studies were not externally preregistered.

4. Use of generative AI

OpenAI GPT-6, GPT-6.1, and Anthropic Claude Opus 5.5 assisted with research ideas, mathematical derivations, programming, simulation execution, analysis, drafting, figures, and editing. The author directed the research question, scope, and restrictions, and required retention of adverse outcomes. Verification used separate mathematical derivations, exact-arithmetic checks, software tests, primary-source checks, and comparisons with saved numerical outputs. The author supplied the study direction and author information. The mathematical and numerical checks included AI assistance.

ACKNOWLEDGMENTS

This work was funded by Emergent Ventures.

CONFLICT OF INTEREST

The author declares no competing interests.

DATA AVAILABILITY

This study uses only synthetic data. Code, study settings, random-generator states, complete present-study decision tables, numerical cutoff certificates, and saved figure and table data are available at <https://github.com/dvgyl/time-reversal-inference>. The versioned archive is deposited at doi:10.5281/zenodo.23223118. The compact release omits the full raw trajectories and detailed per-record numerical files, which are retained locally. The code and settings support regeneration. Floating-point arrays can differ across software versions and platforms. The release supplies the recorded environment and checksums for the complete saved package.

-
- [1] Gonzalez, J. P., Neu, J. C. & Teitworth, S. W. Experimental metrics for detection of detailed balance violation. *Physical Review E* **99**, 022143 (2019).
 - [2] Ohga, N., Ito, S. & Kolchinsky, A. Thermodynamic Bound on the Asymmetry of Cross-Correlations. *Physical Review Letters* **131**, 077101 (2023).
 - [3] Cerasoli, S., Ciliberto, S., Marinari, E., Oshanin, G., Peliti, L. & Rondoni, L. Spectral fingerprints of nonequilibrium dynamics: The case of a Brownian gyrator. *Physical Review E* **106**, 014137 (2022).
 - [4] Seara, D. S., Machta, B. B. & Murrell, M. P. Irreversibility in dynamical phases and transitions. *Nat. Commun.* **12**, 392 (2021).
 - [5] Tong, H. & Zhang, Z. On time-reversibility of multivariate linear processes. *Statistica Sinica* **15**, 495–504 (2005).
 - [6] Bhandari, A. Measuring the Arrow of Time: Identification, Estimation, and Inference for Directional Structure in Multivariate Time Series. Preprint, arXiv:2608.13431v1 (2026).
 - [7] S. Stehbens, H. Pemble, L. Murrow, and T. Wittmann, Imaging intracellular protein dynamics by spinning disk confocal microscopy, *Methods in Enzymology* **504**, 293–313 (2012).
 - [8] K. Berg-Sørensen and H. Flyvbjerg, Power spectrum analysis for optical tweezers, *Review of Scientific Instruments* **75**, 594–612 (2004).
 - [9] C. Battle, C. P. Broedersz, N. Fakhri, V. F. Geyer, J. Howard, C. F. Schmidt, and F. C. MacKintosh, Broken detailed balance at mesoscopic scales in active biological systems, *Science* **352**, 604–607 (2016).
 - [10] F. S. Gnesotto, F. Mura, J. Gladrow, and C. P. Broedersz, Broken detailed balance and non-equilibrium dynamics in living systems: a review, *Reports on Progress in Physics* **81**, 066601 (2018).
 - [11] D. S. Seara, V. Yadav, I. Linsmeier, A. P. Tabatabai, P. W. Oakes, S. M. A. Tabei, S. Banerjee, and M. P. Murrell, Entropy production rate is maximized in non-contractile actomyosin, *Nature Communications* **9**, 4948 (2018).
 - [12] J. Li, J. M. Horowitz, T. R. Gingrich, and N. Fakhri, Quantifying dissipation using fluctuating currents, *Nature Communications* **10**, 1666 (2019).
 - [13] F. Mura, G. Gradziuk, and C. P. Broedersz, Nonequilibrium Scaling Behavior in Driven Soft Biological Assemblies, *Physical Review Letters* **121**, 038002 (2018).
 - [14] G. Gradziuk, F. Mura, and C. P. Broedersz, Scaling behavior of nonequilibrium measures in internally driven elastic assemblies, *Physical Review E* **99**, 052406 (2019).
 - [15] R. Filliger and P. Reimann, Brownian gyrator: A minimal heat engine on the nanoscale, *Physical Review Letters* **99**, 230602 (2007).
 - [16] A. Argun, J. Soni, L. Dabelow, S. Bo, G. Pesce, R. Eichhorn, and G. Volpe, Experimental realization of a minimal microscopic heat engine, *Physical Review E* **96**, 052106 (2017).
 - [17] Hansen, L. P. & Sargent, T. J. The dimensionality of the aliasing problem in models with rational spectral densities. *Econometrica* **51**, 377–387 (1983).
 - [18] Parra, G. & Tobar, F. Spectral mixture kernels for multi-output Gaussian processes. In *Advances in Neural Information Processing Systems* **30** (2017).
 - [19] Rutten, V. M. S., Bernacchia, A., Sahani, M. & Hennequin, G. Non-reversible Gaussian processes for identifying latent dynamical structure in neural data. In *Advances in Neural Information Processing Systems* **33** (2020).
 - [20] R. C. Jennison, A Phase Sensitive Interferometer Technique for the Measurement of the Fourier Transforms of Spatial Brightness Distributions of Small Angular Extent, *Monthly Notices of the Royal Astronomical Society* **118**, 276–284 (1958).
 - [21] Thyagarajan, N., Nityananda, R. & Samuel, J. Invariants in co-polar interferometry: An Abelian gauge theory. *Physical Review D* **105**, 043019 (2022).
 - [22] F. P. Kelly, *Reversibility and Stochastic Networks* (Wiley, Chichester, 1979), Chap. 1.
 - [23] É. Roldán and J. M. R. Parrondo, Estimating Dissipation from Single Stationary Trajectories, *Physical Review Letters* **105**, 150607 (2010).
 - [24] U. Seifert, Stochastic thermodynamics, fluctuation theorems and molecular machines, *Reports on Progress in Physics* **75**, 126001 (2012).
 - [25] Laurent, B. & Massart, P. Adaptive estimation of a quadratic functional by model selection. *Annals of Statistics* **28**, 1302–1338 (2000).
 - [26] Hsu, D., Kakade, S. M. & Zhang, T. A tail inequality for quadratic forms of subgaussian random vectors. *arXiv* 1110.2842v1 (2011).
 - [27] Basu, S. & Michailidis, G. Regularized estimation in sparse high-dimensional time series models. *Annals of Statistics* **43**, 1535–1567 (2015).
 - [28] Dufour, J.-M. Exact tests and confidence sets in linear regressions with autocorrelated errors. *Econometrica* **58**, 475–494 (1990).
 - [29] Dufour, J.-M. & King, M. L. Optimal invariant tests for the autocorrelation coefficient in linear regressions with stationary or nonstationary AR(1) errors. *Journal of Econometrics* **47**, 115–143 (1991).
 - [30] Andrews, D. W. K. Exactly Median-Unbiased Estimation of First Order Autoregressive/Unit Root Models. *Econometrica* **61**, 139–165 (1993).
 - [31] Bi, Y. & Calhoun, V. D. Cross-Spectral Witness for Hidden Nonequilibrium Beyond the Scalar Ceiling. Preprint, arXiv:2604.03775v3 (2026).
 - [32] Lucente, D., Baldassarri, A., Puglisi, A., Vulpiani, A. & Viale, M. Inference of time irreversibility from incomplete information: Linear systems and its pitfalls. *Physical Review Research* **4**, 043103 (2022).
 - [33] H. R. Künsch, The jackknife and the bootstrap for general stationary observations, *Annals of Statistics* **17**, 1217–1241 (1989).
 - [34] Nolte, G., Bai, O., Wheaton, L., Mari, Z., Vorbach, S. & Hallett, M. Identifying true brain interaction from EEG data using the imaginary part of coherency. *Clinical Neurophysiology* **115**, 2292–2307 (2004).
 - [35] Clopper, C. J. & Pearson, E. S. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika* **26**, 404–413 (1934).
 - [36] C. R. Harris *et al.*, Array programming with NumPy, *Nature* **585**, 357–362 (2020).
 - [37] P. Virtanen *et al.*, SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python, *Nature Methods* **17**, 261–272 (2020).

- [38] J. D. Hunter, Matplotlib: A 2D graphics environment, *Computing in Science & Engineering* **9**, 90–95 (2007).